

where the bytes live

digitalisierung und programmierung · prof. dr. nicolas meseth

- 1 a file is a sequence of bytes
- 2 where the bytes live
- 3 on your link

part 1

a file is a sequence of bytes

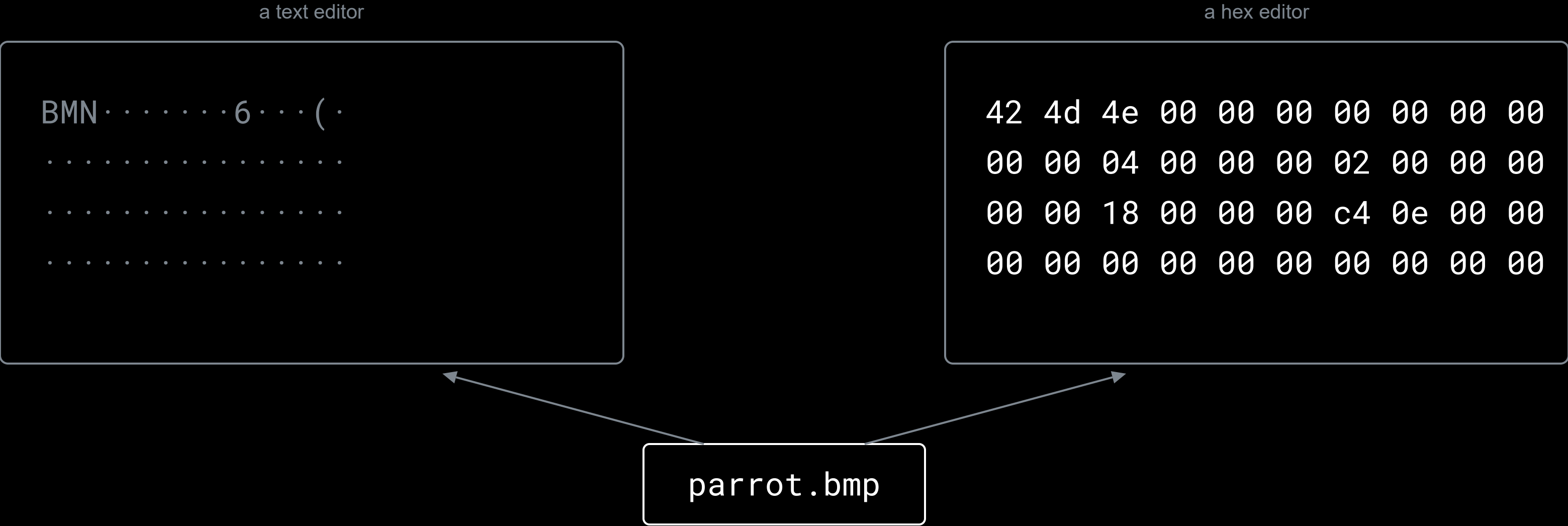
this photo is
196 662 bytes.
where are they
right now?



image: ai-generated (gpt-image-2.5-flare)

a file is a sequence of bytes

the same file, two programs



the bytes do not say what they mean. the program decides how to read them.

78 bytes, all of them

0000	42	4d	4e	00	00	00	00	00	00	00	36	00	00	00	28	00
0010	00	00	04	00	00	00	02	00	00	00	01	00	18	00	00	00
0020	00	00	18	00	00	00	c4	0e	00	00	c4	0e	00	00	00	00
0030	00	00	00	00	00	00	00	00	00	00	ff	ff	ff	ff	00	ff
0040	00	ff	00	00	ff	00	ff	00	ff	00	00	ff	ff	ff		

file header

14 bytes: "bm", the size of the file, where the pixels start

78 bytes, all of them

0000	42	4d	4e	00	00	00	00	00	00	00	36	00	00	00	28	00
0010	00	00	04	00	00	00	02	00	00	00	01	00	18	00	00	00
0020	00	00	18	00	00	00	c4	0e	00	00	c4	0e	00	00	00	00
0030	00	00	00	00	00	00	00	00	00	00	ff	ff	ff	ff	00	ff
0040	00	ff	00	00	ff	00	ff	00	ff	00	00	ff	ff	ff		

info header

40 bytes: width, height, colour depth, and more

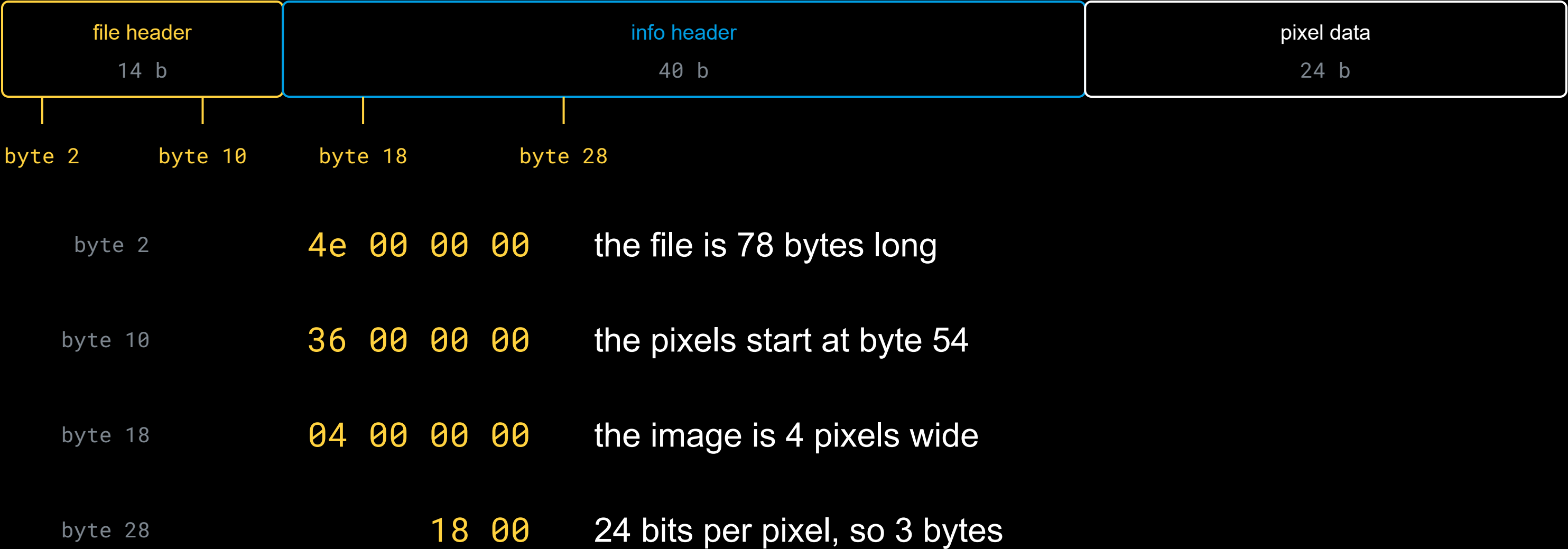
78 bytes, all of them

0000	42	4d	4e	00	00	00	00	00	00	00	36	00	00	00	28	00
0010	00	00	04	00	00	00	02	00	00	00	01	00	18	00	00	00
0020	00	00	18	00	00	00	c4	0e	00	00	c4	0e	00	00	00	00
0030	00	00	00	00	00	00	00	00	00	00	ff	ff	ff	ff	00	ff
0040	00	ff	00	00	ff	00	ff	00	ff	00	00	ff	ff	ff		

pixel data

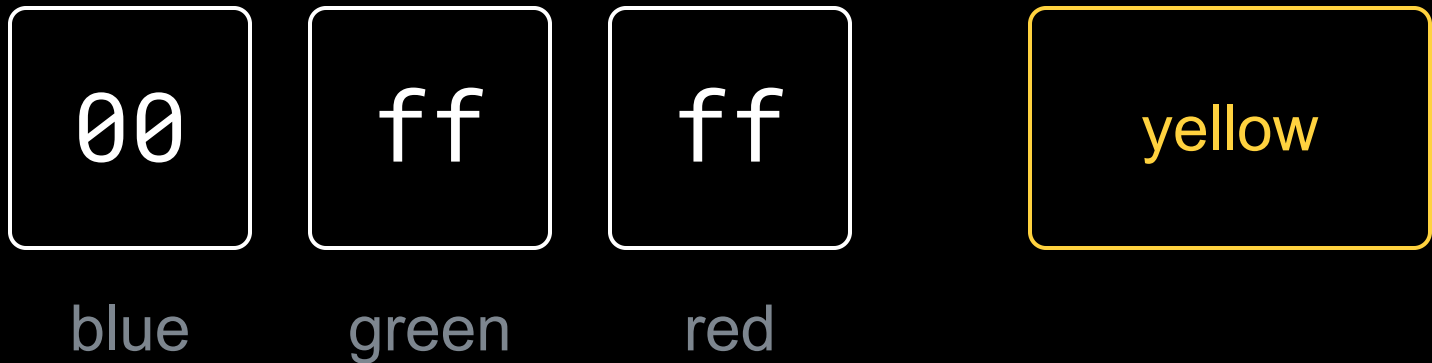
24 bytes: 4 × 2 pixels × 3 bytes

the header says how to read the rest

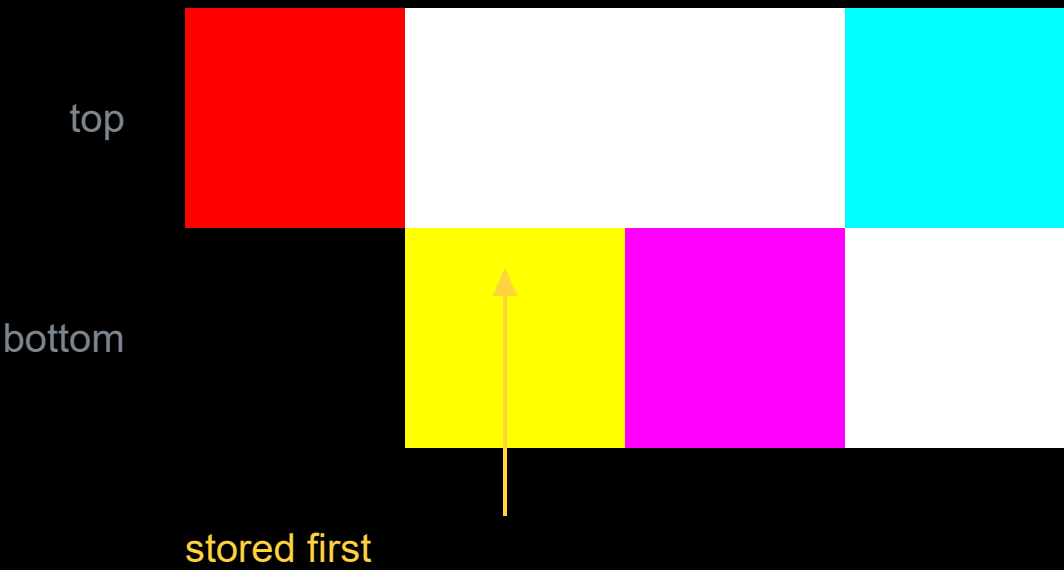


$4 \times 2 \times 3 = 24$ bytes. the program knows that before it has seen a single pixel.

two decisions somebody made in 1990



not red, green, blue. the other way round.



bitmaps stand upside down in the file.

not cyan, and not the top row. yellow, at the bottom.

how does a program know what it is looking at?

the extension

.bmp, in the name, not
in the file.
rename it and the bytes
do not change.

magic bytes

BM at the front, in the
file.

png has eight, pdf has
%PDF. a text file has
none.

a type field

your frame, in front of
the data.

the class decides which
number means which
type.

try it: inside a file

picture:



pixels:

8 × 8

16 × 16

32 × 32

what a viewer makes of the bytes



click a pixel

open as text

reset file

the bytes (a hex editor)

0000	42	4d	36	03	00	00	00	00	00	00	36	00	00	00	28	00
0010	00	00	10	00	00	00	10	00	00	00	01	00	18	00	00	00
0020	00	00	00	03	00	00	c4	0e	00	00	c4	0e	00	00	00	00
0030	00	00	00	00	00	00	0a	0b	17	0e	11	1c	6b	69	62	45
0040	41	41	11	16	1e	12	19	20	13	19	20	10	18	1e	10	1a
0050	44	1b	22	42	0d	17	82	19	21	b0	1b	25	e4	26	3a	f6
0060	28	3d	f9	23	36	f9	0c	0d	1a	56	5e	66	a2	a4	a1	38
0070	3a	3e	10	17	1e	12	1c	21	1d	27	31	11	13	21	0b	0b
0080	1d	06	06	18	2e	33	62	28	30	91	1b	1e	da	1e	26	e6
0090	26	3a	f2	27	37	f5	21	25	32	9e	a8	ae	a2	a9	ab	55
00a0	57	55	16	17	1a	31	2e	31	3a	33	36	2a	24	27	25	20
00b0	23	1b	16	1f	40	3a	45	a2	94	a8	4b	4d	c6	18	12	e0
00c0	17	0c	df	12	0b	da	50	57	62	9e	ab	b6	a1	a9	af	6a
00d0	69	68	12	10	11	4d	48	49	2d	27	26	2e	27	27	24	1e
00e0	20	25	1f	23	96	89	8e	c5	b5	bf	97	90	c5	22	29	ee
00f0	1d	18	eb	1a	13	e9	83	8f	9c	9d	ad	bb	a2	ad	b5	90
0100	8e	88	42	3a	39	4d	45	43	2c	26	25	2c	25	25	2d	26
0110	26	75	6b	6e	ca	bb	be	ce	bf	c5	ca	bd	c7	72	7f	de
0120	1f	28	f2	1e	24	f2	00	c7	b0	0e	eb	bd	08	e0	b7	0e

hover a byte

☐ file header ☐ info header ☐ pixel data ☒ selected pixel

822 bytes = 14 file header + 40 info header + 768 pixel data (16 × 16 pixels × 3 bytes)

photos: ai-generated (gpt-image-2)

what a file is

a sequence of bytes, plus an agreement about how to read it.

the agreement is written down, not natural.

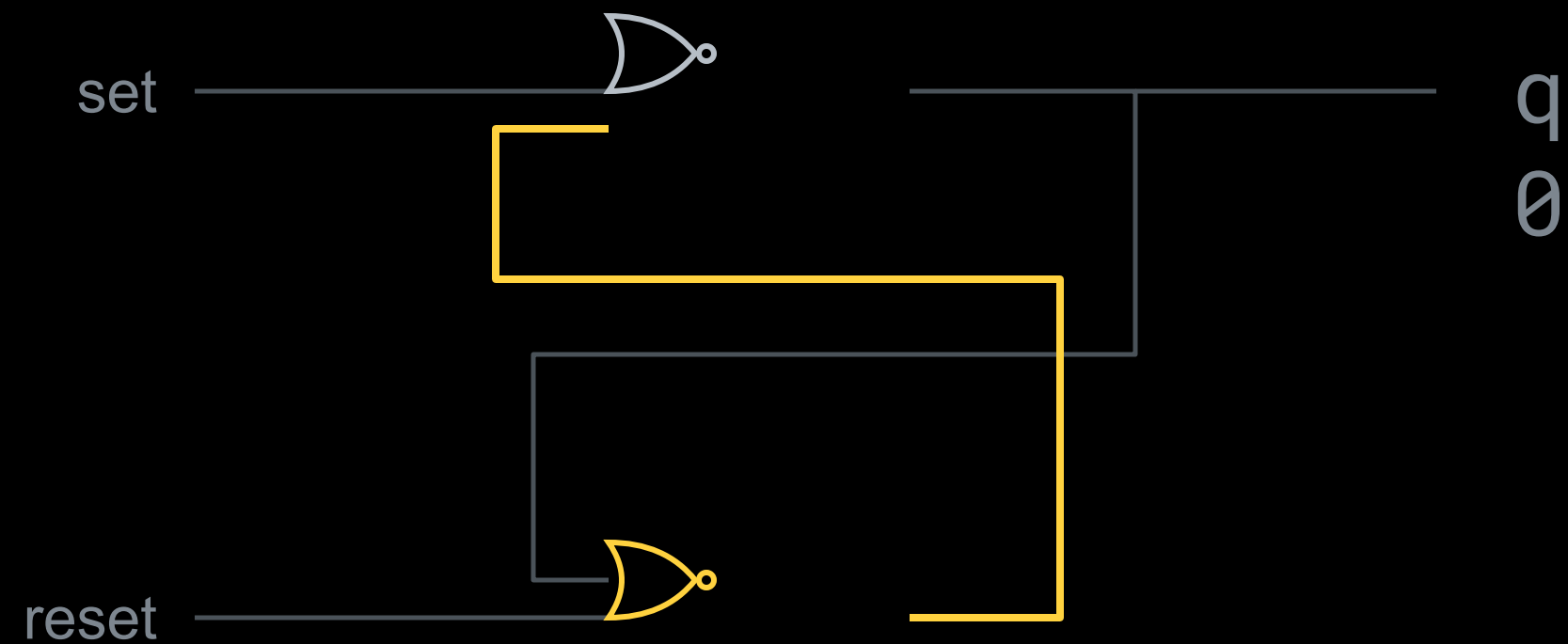
a broken byte in the data is a small injury. in the header it is a large one.

every file on your laptop works this way. yours will too.

part 2

where the bytes live

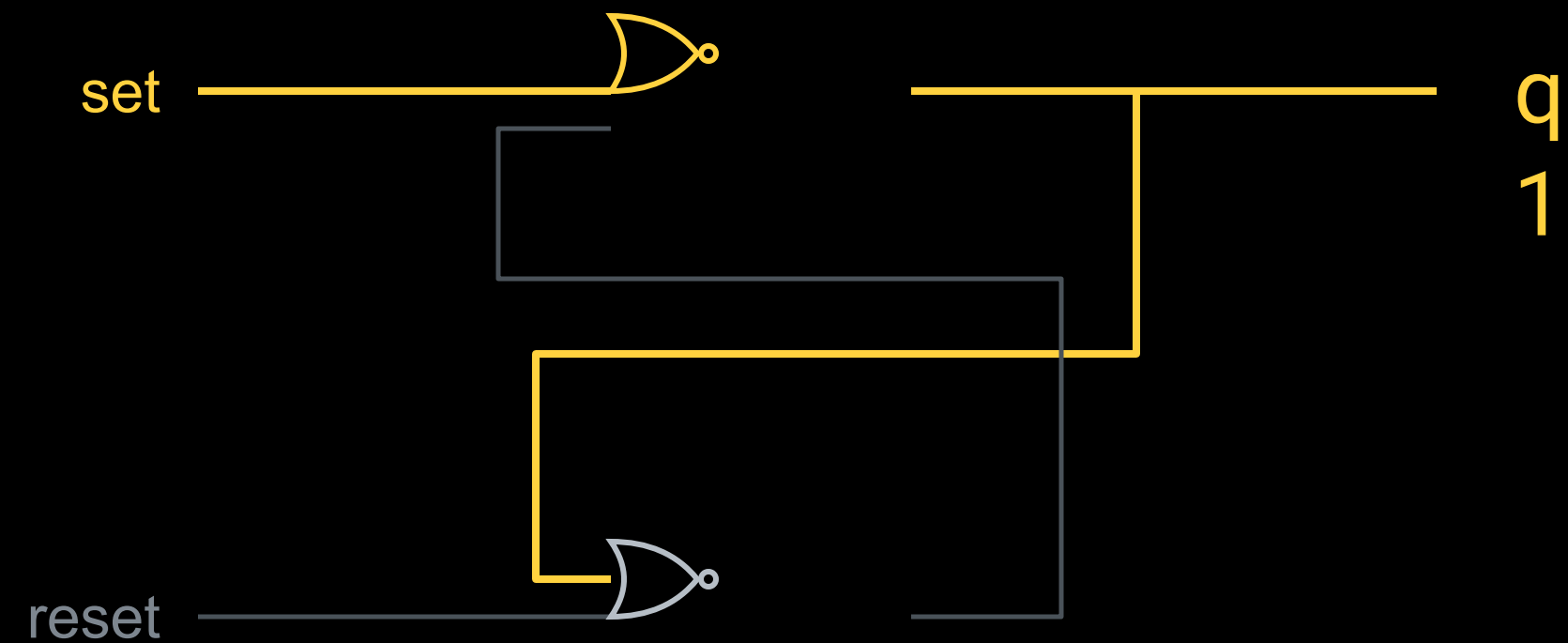
a bit that remembers



nothing happens. q is 0.

the same gates that add and compare. wired in a circle, they remember.

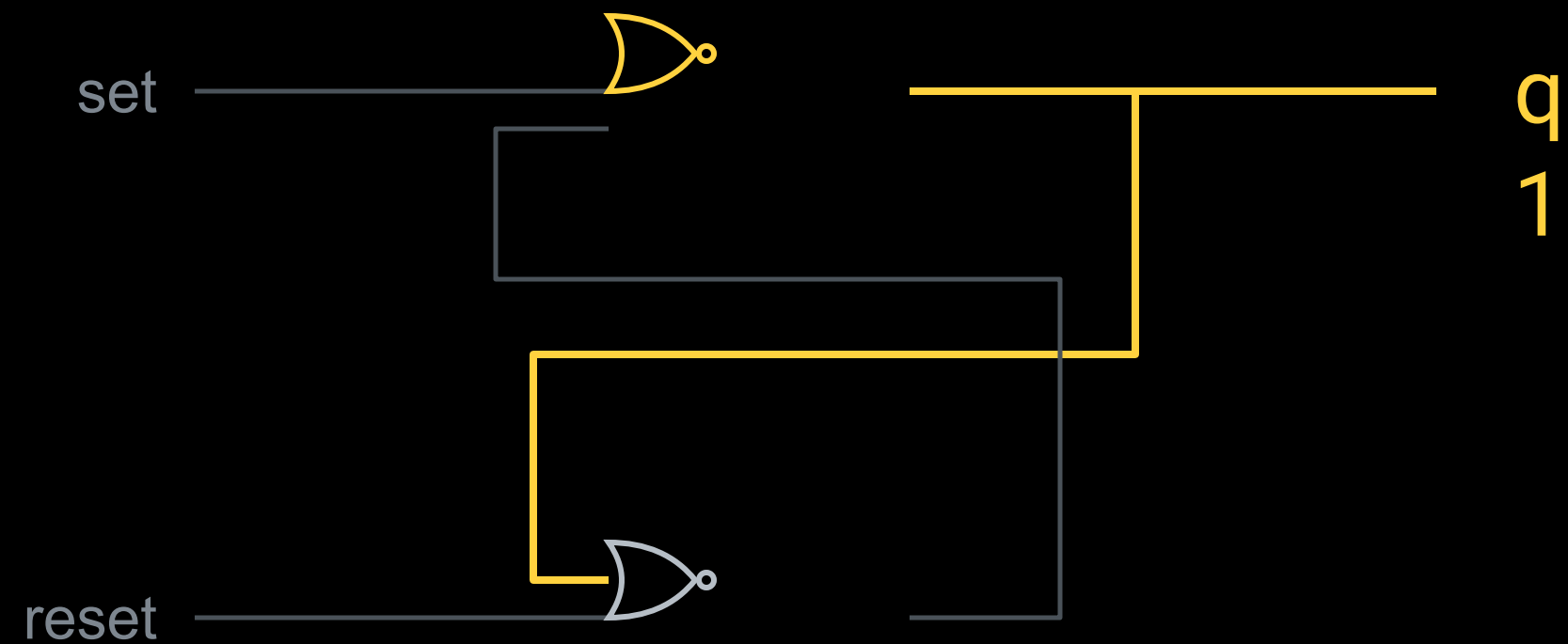
a bit that remembers



a short pulse on set, and q becomes 1.

the same gates that add and compare. wired in a circle, they remember.

a bit that remembers



set is off again, and q stays 1. the loop feeds itself.

the same gates that add and compare. wired in a circle, they remember.

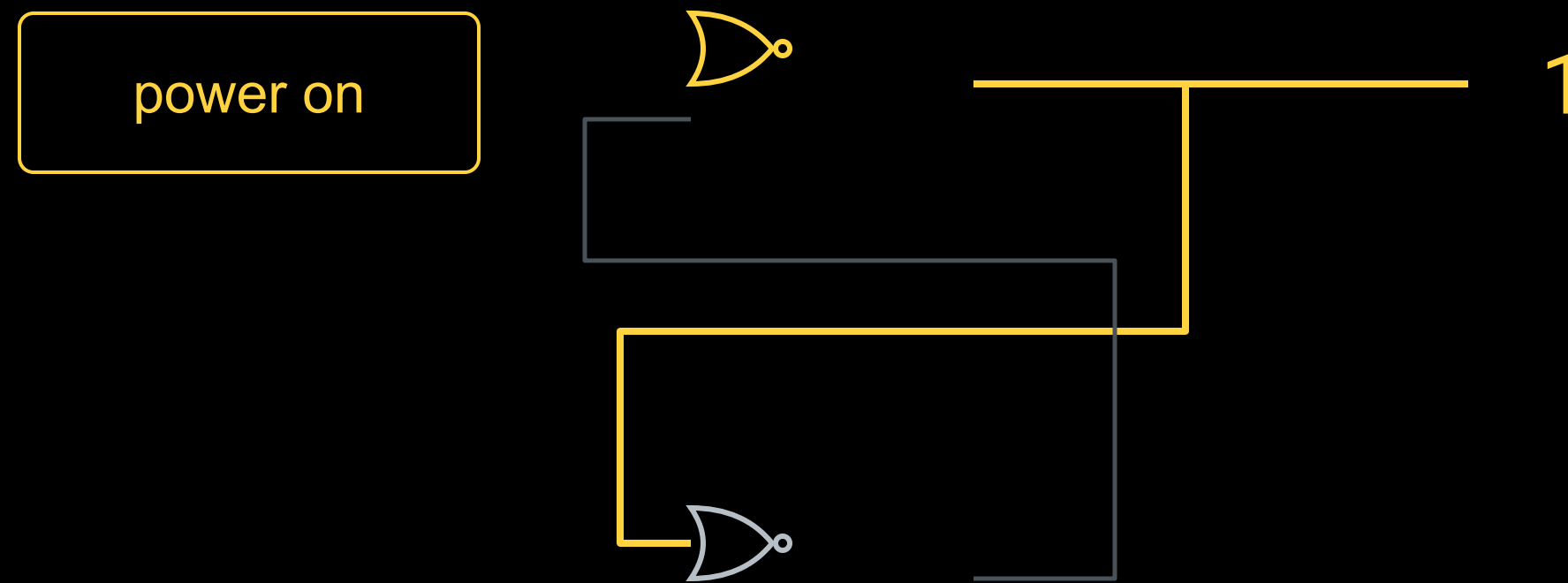
millions of them, each with a number



billions of rows, all alike. the number is the only handle.

a program never says "the number over there". it says "the byte at 4711".

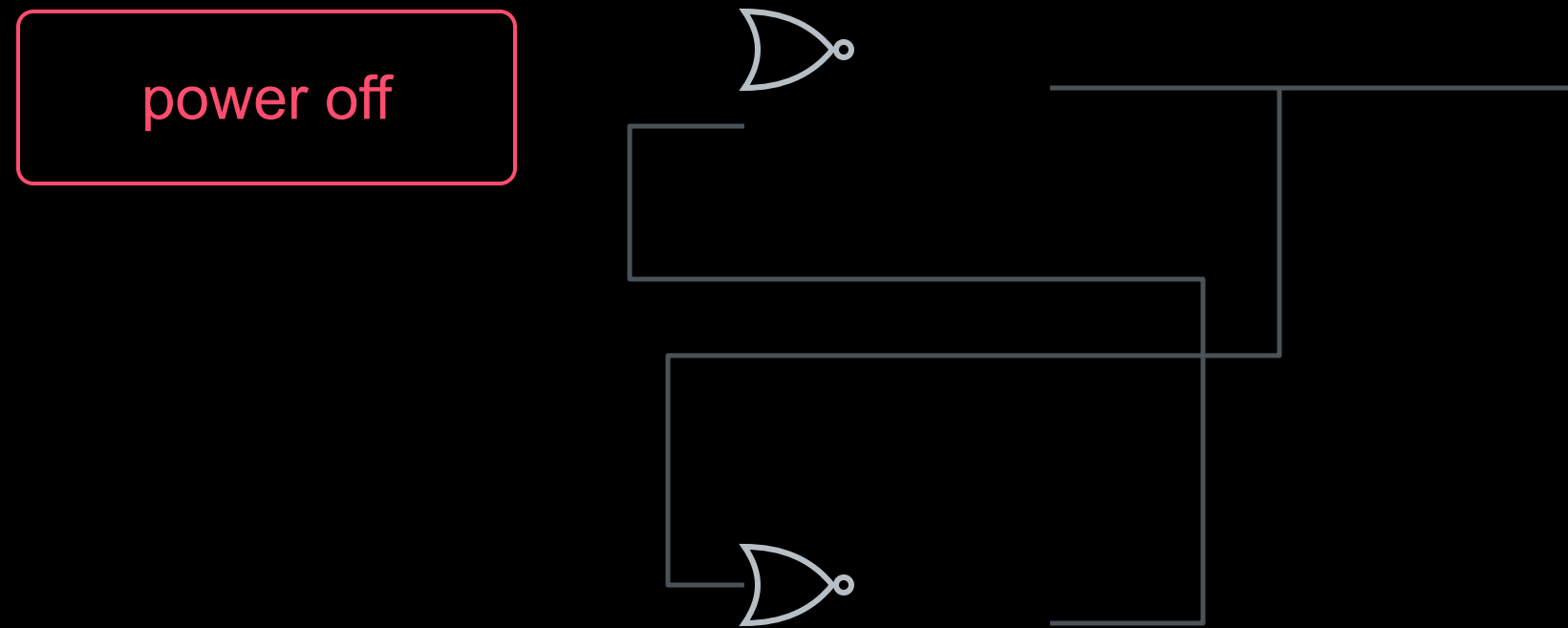
pull the plug



the two gates keep feeding each other.

ram answers in nanoseconds because nothing has to move. that is what it costs.

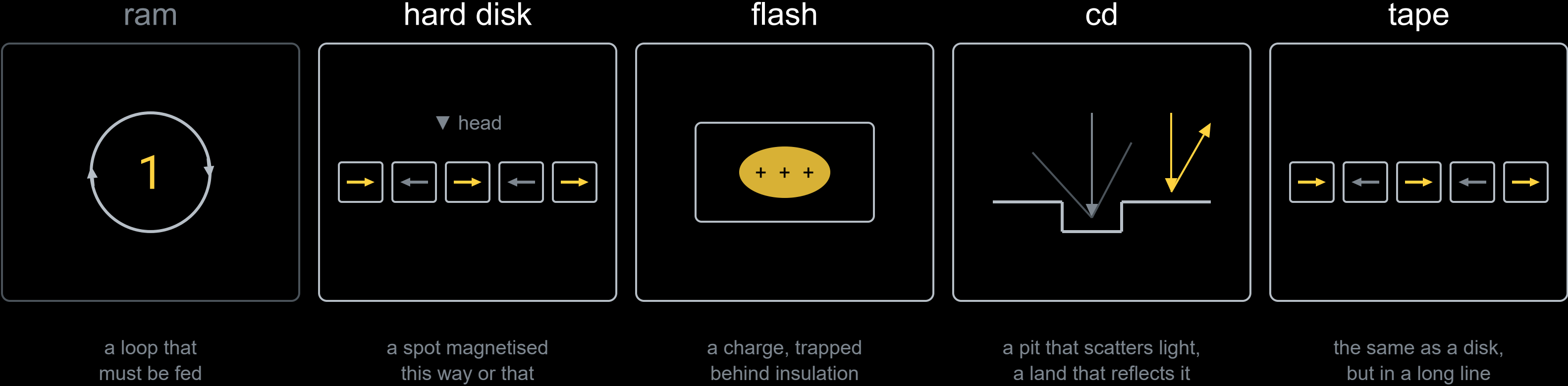
pull the plug



the loop collapses. nobody deleted anything.

ram answers in nanoseconds because nothing has to move. that is what it costs.

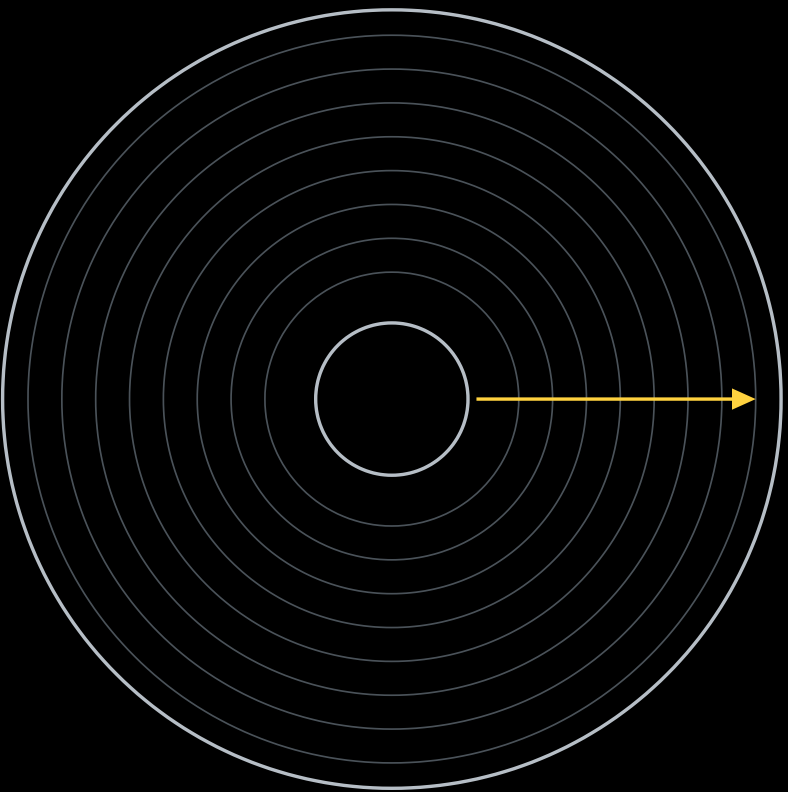
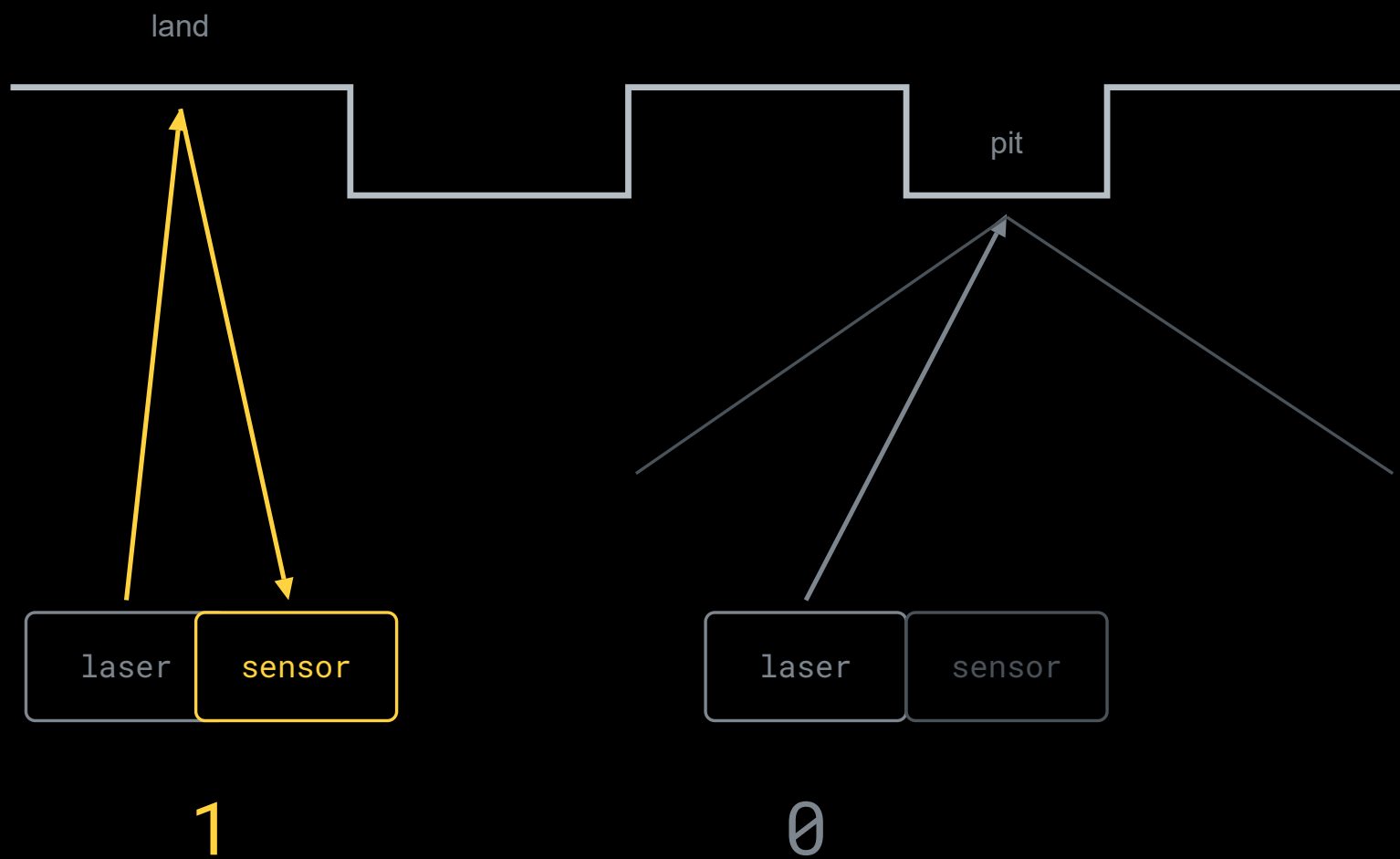
what is a bit made of here?



only the first one needs power. and none of them knows whether its bit belongs to a parrot.

five materials, one job: hold a state that stays put, and answer when asked.

a cd reads bits with light



one spiral, from the inside out

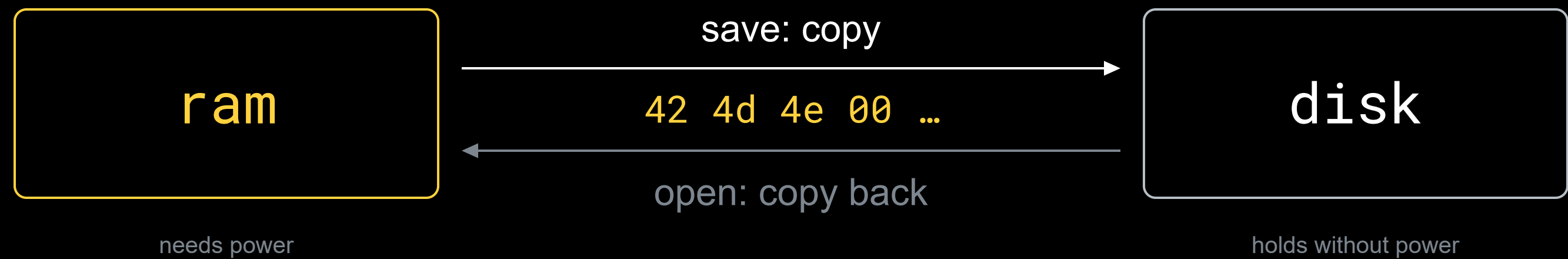
a laser from below, and a reflection that comes back or does not.

a lamp, a sensor, and a reflection that is there or is not. that is your device, on a spiral.

the medium outlives its reader

a floppy disk holds its bits fine. finding a drive is the problem.
tape is still how archives keep petabytes, for decades.
a state without the agreement that reads it is not information yet.

what saving means



copied, not moved: after saving, the bytes are in both places, and they are the same bytes.

copied, not moved. the same sequence of bytes, in a place that needs no power.

four places, one sequence

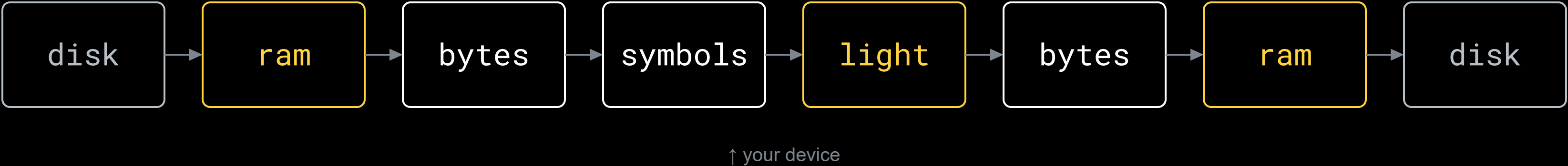
	what a bit is	how long to reach it	without power
register	eight flip-flops in the processor	under a nanosecond	gone
main memory	a cell with an address	nanoseconds	gone
disk, flash	a magnetised spot, a trapped charge	milliseconds	holds
cd, tape	a pressed pit, a magnetised stripe	seconds to find the place	holds for decades

the faster it answers, the more it depends on the power staying on.

part 3

on your link

the whole way



at no point is it a picture. it is the same sequence of bytes, in different places.

storing, representing, transferring, processing: the map of this module, read from below.

what this means for challenge 4

read the file as bytes, not as a picture.

send the header too. without it the other side cannot read anything.

"it displays" proves nothing. the hash decides.

and one more: a byte you change in the header costs you the whole file.



a magnetised spot.
that is your photo.